

Lec 9-2, Nov 5, 2025

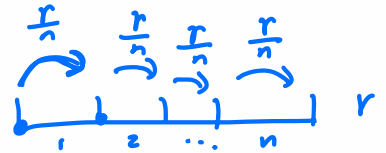
- General powers and general exponential functions (7.3)
- Indeterminate forms (7.5)
- L' Hospital's rule (7.5)
- Apply L' Hospital's rule *carefully* !
- Limits of products and quotients (Not in Thomas')
- Relative rates of growth (7.8)
- Big-oh and little-oh notations (7.8)
- Proof of L' Hospital's rule (optional)
 - Cauchy's mean value theorem (optional)

Exercise. Suppose that P_0 dollars are deposited into an account with annual interest rate r , compounded n times a year. When n is very large, approximate the balance (i.e., the amount of \$) in the account t years after the deposit using a limit. (i.e., $n \rightarrow \infty$).

Sol: For each period, if the balance at the beginning of the period is denoted by x , then the balance after the period is

$$x + (\text{interest}) x \cdot r \cdot \frac{1}{n} = x \left(1 + \frac{r}{n}\right)$$

In t years, we have $t \cdot n$ periods.



Hence, the balance at the end of t years is



$$P_0 \cdot \underbrace{\left(1 + \frac{r}{n}\right) \cdot \left(1 + \frac{r}{n}\right) \cdot \dots \cdot \left(1 + \frac{r}{n}\right)}_{t \cdot n \text{ periods}} = P_0 \left(1 + \frac{r}{n}\right)^{t \cdot n}$$

Compute:

$$\lim_{n \rightarrow \infty} P_0 \left(1 + \frac{r}{n}\right)^{t \cdot n} \quad \begin{matrix} u = \frac{r}{n} \rightarrow 0 \\ n = \frac{r}{u} \end{matrix} \quad \lim_{u \rightarrow 0} P_0 (1+u)^{\frac{t \cdot r}{u}} = P_0 \lim_{u \rightarrow 0} \left(1+u\right)^{\frac{1}{u}}^{t \cdot r}$$

Let $g(x) = x^{t \cdot r}$, $x > 0$. $\Rightarrow g$ is continuous on $(0, \infty)$

$$\begin{aligned}\therefore \lim_{n \rightarrow \infty} P_0 \left(1 + \frac{1}{n}r\right)^{tn} &= P_0 \lim_{u \rightarrow 0} \left((1+u)^{\frac{1}{u}}\right)^{t \cdot r} \\ &= P_0 \lim_{u \rightarrow 0} g\left((1+u)^{\frac{1}{u}}\right) \\ &\because g \text{ cont.} \quad P_0 \cdot g\left(\lim_{u \rightarrow 0} (1+u)^{\frac{1}{u}}\right) \\ &= P_0 \cdot g(e) \\ &= P_0 \cdot e^{t \cdot r}\end{aligned}$$

At the end of t years, you will get $P_0 \cdot e^{tr}$ dollars.



General logarithmic functions

- Fix $a \in (0, \infty) \setminus \{1\}$.

(or, one-to-one)

- The function f given by $f(x) = a^x$ is monotonic, so it is injective on domain $\mathbb{R}$.

- From the definition $a^x = e^{x \ln a}$ and the fact that $\text{range}(\exp) = (0, \infty)$, it follows that the range of f is also $(0, \infty)$.

- Hence, $f: \mathbb{R} \rightarrow (0, \infty)$ has an inverse function $\log_a: (0, \infty) \rightarrow \mathbb{R}$

called the logarithmic function with base a .

- $x_0 = a^{y_0} \iff y_0 = \log_a(x_0)$

- On the other hand,

$$x_0 = a^{y_0} \iff x_0 = e^{y_0 \ln a} \iff y_0 \ln a = \ln x_0$$

$$\iff y_0 = \frac{\ln x_0}{\ln a}$$

Hence,

$$\log_a x = \frac{\ln x}{\ln a} \quad \left(= \frac{\log_e x}{\log_e a} \right) \quad (*)$$

e.g. $[\log_{10}(x)]' = \frac{1}{x} \cdot \frac{1}{\ln 10}$

$\Uparrow$

$$\log_{10}(x) = \frac{\ln x}{\ln 10}$$

e.g. $\log_{10} 100 = 2$

$\log_{10} 10000 = 4$

$\log_e x = \ln x$

e.g. $(\ln x)' = \frac{1}{x}$

• With the equality (*) above, one can show that

the four algebraic properties for $\ln$ also hold for $\log_a x$.

e.g., $\log_a(xy) = \frac{\ln(xy)}{\ln a} = \frac{\ln x + \ln y}{\ln a} = \frac{\ln x}{\ln a} + \frac{\ln y}{\ln a} = \log_a(x) + \log_a(y)$

$$\ln(xy) = \ln x + \ln y, \quad \forall x > 0, y > 0$$

• From the equality (*), we can see that

$$\frac{d}{dx} \log_a x = \frac{1}{x \ln a}$$

$\frac{d}{dx} \left(\frac{\ln x}{\ln a} \right)$

Indeterminate forms

- Consider $\lim_{x \rightarrow 0} \frac{\sin x}{x}$: it has the form " $\frac{0}{0}$ " by limit laws, while the limit is 1.

- Consider $\lim_{x \rightarrow 0} \frac{kx}{x}$: it also has the form " $\frac{0}{0}$ " by limit laws, while the limit is k .

[where k is a finite constant].

- These above are examples of an indeterminate form.

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$$

↑
Limit law

- Roughly speaking, an indeterminate form is a form of "limit"

which does not have a fixed value and cannot be determined by the limit laws.

- We need to use other methods to find the limit value.

- There are 7 indeterminate forms:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty, 0^0, 1^\infty, \infty^0.$$

e.g. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$, then we cannot say anything

about $\lim_{x \rightarrow a} f(x)^{g(x)}$. [This is " 0^0 "]

e.g. (a) Let $f(x) \equiv 0$, $g(x) = x^2$. $\lim_{x \rightarrow 0} f(x)^{g(x)} = \lim_{x \rightarrow 0} 0^{(x^2)} = \lim_{x \rightarrow 0} 0 = 0$.

(b) Let $f(x) = x$, $g(x) \equiv 0$. $\lim_{x \rightarrow 0} f(x)^{g(x)} = \lim_{x \rightarrow 0} x^0 = \lim_{x \rightarrow 0} 1 = 1$.

e.g. 1) " $\frac{0}{0}$ ": as $x \rightarrow 0$, $\frac{\sin x}{x} \rightarrow 1$, $\frac{2x}{x} \rightarrow 2$.

2) " $\frac{\infty}{\infty}$ ": as $x \rightarrow \infty$, $\frac{2x}{x} \rightarrow 2$, $\frac{3x}{x} \rightarrow 3$.

$$3) 0 \cdot \infty : \text{as } x \rightarrow \infty, \frac{1}{x} \cdot x \rightarrow 1, \frac{1}{x} \cdot x^2 \rightarrow \infty.$$

$$4) \infty - \infty : \text{as } x \rightarrow \infty, x - x \rightarrow 0, 2x - x \rightarrow \infty, x - x^2 \rightarrow -\infty.$$

$$5) 0^0 : \text{as } x \rightarrow \infty, (e^{-x})^{\frac{1}{x}} \rightarrow e^{-1}, (e^{-x})^{\frac{2}{x}} \rightarrow e^{-2}.$$

$$6) 1^\infty : \text{as } x \rightarrow 0^+, (1+x)^{\frac{1}{x}} \rightarrow e, (1+x)^{\frac{2}{x}} \rightarrow e^2.$$

$$7) \infty^0 : \text{as } x \rightarrow 0^+, (e^{\frac{1}{x}})^x \rightarrow e, (e^{\frac{1}{x}})^{2x} \rightarrow e^2.$$

Q: What about 0^∞ ? Is it an indeterminate form?

A: No. Assume good conditions. If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$,

$$\text{then } \lim_{x \rightarrow a} f(x)^{g(x)} \stackrel{f(x) > 0}{=} \lim_{x \rightarrow a} e^{g(x) \ln f(x)}$$

$$\stackrel{\text{exp cont'}}{=} e^{\lim_{x \rightarrow a} [g(x) \ln f(x)]}$$

$$= e^{\lim_{x \rightarrow a} g(x) \cdot \lim_{x \rightarrow a} \ln f(x)}$$

$$(\approx e^{\infty \cdot (-\infty)})$$

$$\approx e^{-\infty}$$

$$= 0.$$

$$\left(\lim_{x \rightarrow a} g(x) = \infty \right.$$

$$\left. \lim_{x \rightarrow a} f(x) = 0 \right.$$

$$\ln(\lim_{x \rightarrow a} f(x)) = -\infty \quad \square.$$

$$\lim_{x \rightarrow a} g(x) \cdot \ln(\lim_{x \rightarrow a} f(x)) = -\infty$$

One method that is very effective in finding limits of indeterminate forms

is L'Hôpital's rule!

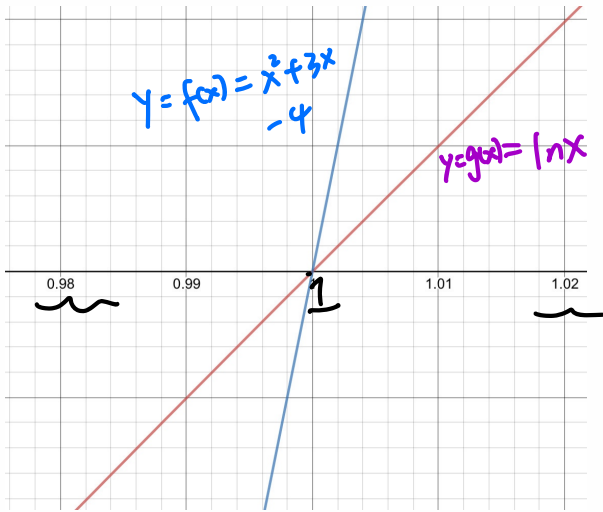
$$0.1^{99} = \dots$$

$$0.01^{999} = \dots$$

$$0.001^{9999} = \dots$$

• L'Hôpital's Rule (or L'Hospital's Rule)

— Consider $\lim_{x \rightarrow 1} \frac{x^2+3x-4}{\ln x}$ It is an indeterminate form " $\frac{0}{0}$ ".



• As $x \rightarrow 1$, x^2+3x-4 and $\ln x$ locally look like their tangent lines at $x=1$.

$$(f(x) \approx f(1) + f'(1)(x-1),$$

$$g(x) \approx g(1) + g'(1)(x-1) \text{ locally at } x=1)$$

$$\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} \approx \lim_{x \rightarrow 1} \frac{\overset{=0}{f(1)} + \cancel{f'(1)(x-1)}}{\underset{=0}{g(1)} + \cancel{g'(1)(x-1)}} \quad \text{or} \quad \lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)}$$

— Consider $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$, where we suppose f and g both are continuously

differentiable on $(c-a, c+a)$ [a function is continuously differentiable if its

derivative function is continuous] and $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0 = f(c) = g(c)$.

— Then

$$\begin{aligned} \lim_{x \rightarrow c} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{g(x) - g(c)} = \lim_{x \rightarrow c} \frac{\frac{f(x) - f(c)}{x - c}}{\frac{g(x) - g(c)}{x - c}} \stackrel{\text{limit laws}}{=} \frac{\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}}{\lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c}} \\ &\stackrel{\substack{\uparrow \\ f(c)=0 \\ g(c)=0}}{=} \frac{f'(c)}{g'(c)} = \frac{\lim_{x \rightarrow c} f'(x)}{\lim_{x \rightarrow c} g'(x)} \stackrel{\substack{\uparrow \\ \text{limit laws}}}{=} \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \end{aligned}$$

f, g diff' and $g'(c) \neq 0$.
 f', g' are continuous

The general form of this is now known as L'Hôpital's rule, which requires weaker assumptions and covers a lot of cases.

★ Theorem (L'Hôpital's Rule, or L'Hospital's Rule)

Let $c \in \mathbb{R}$. Suppose that f and g are differentiable on $D := (c - a, c + a) \setminus \{c\}$ for some $a > 0$, and that $g'(x) \neq 0$ for all $x \in D$. Suppose that one of the following two conditions holds:

- ▶ $\lim_{x \rightarrow c} f(x) = 0 = \lim_{x \rightarrow c} g(x)$; "0/0"
- ▶ $\lim_{x \rightarrow c} f(x) \in \{-\infty, \infty\}$ and $\lim_{x \rightarrow c} g(x) \in \{-\infty, \infty\}$. $\frac{\infty}{\infty}, \frac{-\infty}{\infty}, \frac{\infty}{-\infty}, \frac{-\infty}{-\infty}$

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)},$$

provided that the limit on the right exists (or is ∞ or $-\infty$).

Historical remark: The L'Hôpital rule was first discovered by Johann Bernoulli, who was Euler's mentor and the younger brother of Jacob Bernoulli who found $e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$.

Johann Bernoulli

Johann Bernoulli (1667–1748) was born in Basel, Switzerland. Johann worked for a year in his father's spice business, but he was not a success. He enrolled in Basel University to study medicine, but his brother Jacob, twelve years older and a Professor of Mathematics, led him into mathematics. Together, they studied the papers of Leibniz on the new subject of calculus. Johann received his doctorate at Basel University and joined the faculty at Groningen in Holland, but upon Jacob's death in 1705, he returned to Basel and was awarded Jacob's chair in mathematics. Because of his many advances in the subject, Johann is regarded as one of the founders of calculus.



While in Paris in 1692, Johann met the Marquis Guillaume Francois de L'Hospital and agreed to a financial arrangement under which he would teach the new calculus to L'Hospital, giving L'Hospital the right to use Bernoulli's lessons as he pleased. This was subsequently continued through a series of letters. In 1696, the first book on differential calculus, *L'Analyse des Infiniment Petits*, was published by L'Hospital. Though L'Hospital's name was not on the title page, his portrait was on the frontispiece and the preface states "I am indebted to the clarifications of the brothers Bernoulli, especially the younger." The book contains a theorem on limits later known as L'Hospital's Rule although it was in fact discovered by Johann Bernoulli. In 1922, manuscripts were discovered that confirmed the book consisted mainly of Bernoulli's lessons. And in 1955, the L'Hospital-Bernoulli correspondence was published in Germany.

Analyse
→ of
infinite
→ small

Every year, $\approx 300 + 200$ libra
 ≈ 227 Kg silver
 $\approx 227 \times 5751$ RMB
 ≈ 1.3 M RMB

Remark.

- (i) L'Hôpital's rule is also valid if we replace " $\lim_{x \rightarrow c}$ " with " $\lim_{x \rightarrow c^+}$ " or " $\lim_{x \rightarrow c^-}$ ".
- (ii) If D is changed to an unbounded interval, then L'Hôpital's rule is also valid if we replace " $\lim_{x \rightarrow c}$ " with " $\lim_{x \rightarrow \infty}$ " or " $\lim_{x \rightarrow -\infty}$ ".

e.g. (a) $\lim_{x \rightarrow 1} \frac{x^3 + 3x - 4}{x^4 - 6x^2 + 5} = ?$

Sol': • we check: $\lim_{x \rightarrow 1} x^3 + 3x - 4 = 0$,
 $\lim_{x \rightarrow 1} x^4 - 6x^2 + 5 = 0$. $\therefore$ It is " $\frac{0}{0}$ " and L' rule holds.

$\therefore$ LHS $\stackrel{\text{L' rule}}{=} \lim_{x \rightarrow 1} \frac{3x^2 + 3}{4x^3 - 12x} = \lim_{x \rightarrow 1} \frac{3x^2 + 3}{4x^3 - 12x} = -\frac{3}{4}$. [2]

(b) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = ?$

Sol': we check: $\lim_{x \rightarrow 0} x - \sin x = 0$, $\lim_{x \rightarrow 0} x^3 = 0$.

$\therefore$ LHS $\stackrel{\text{L' rule}}{=} \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} \stackrel{\text{L' rule}}{=} \lim_{x \rightarrow 0} \frac{\sin x}{6x} \stackrel{\text{L' rule}}{=} \lim_{x \rightarrow 0} \frac{\cos x}{6} = \frac{1}{6}$. [6]

$$(c) \lim_{x \rightarrow 0} \frac{[\ln(x+1)]^3}{e^x - x - 1} = ?$$

$$\text{Sol}': \lim_{x \rightarrow 0} [\ln(x+1)]^3 = 0$$

$$\lim_{x \rightarrow 0} [e^x - x - 1] = 0$$

$\therefore$ L'Hospital's rule holds. ("0/0").

$$\therefore \text{LHS} \stackrel{L'}{=} \lim_{x \rightarrow 0} \frac{3(\ln(x+1))^2 \cdot \frac{1}{x+1}}{e^x - 1} = \lim_{x \rightarrow 0} \frac{3(\ln(x+1))^2}{(x+1)(e^x - 1)}$$

$$\stackrel{L'}{=} \lim_{x \rightarrow 0} \frac{6 \ln(x+1) \cdot \frac{1}{x+1}}{e^x - 1 + (x+1)e^x} = \frac{\lim_{x \rightarrow 0} 6 \ln(x+1) \cdot \frac{1}{x+1}}{\lim_{x \rightarrow 0} e^x - 1 + (x+1)e^x}$$

$$= \frac{6 \cdot 0 \cdot 1}{1} = 0. \quad \square$$

$$(d) \lim_{x \rightarrow 2} \frac{\sin(\pi x)}{(x-2)^2} = ?$$

$$\text{Sol}': \text{Check: } \lim_{x \rightarrow 2+} \sin(\pi x) = 0 = \lim_{x \rightarrow 2+} (x-2)^2$$

$\Rightarrow$ L' rule holds ("0/0")

$$\therefore \lim_{x \rightarrow 2+} \frac{\sin(\pi x)}{(x-2)^2} = \lim_{x \rightarrow 2+} \frac{\pi \cos(\pi x)}{2(x-2)} = \infty$$

Similarly,

$$\lim_{x \rightarrow 2-} \frac{\sin \pi x}{(x-2)^2} = \lim_{x \rightarrow 2-} \frac{\pi \cos \pi x}{2(x-2)} \xrightarrow{\pi} -\infty$$

$\therefore$ The limit of LHS Does Not Exist. □

In (d), we should take the left-hand limit and the right-hand limit separately. By L'Hôpital's rule, it follows that the left-hand limit is $-\infty$ while the right-hand limit is ∞ , so the limit does not exist (in $\mathbb{R} \cup \{\infty, -\infty\}$).

$$(e) \lim_{x \rightarrow \infty} \frac{\ln x}{x^\pi} = ?$$

Sol': $\lim_{x \rightarrow \infty} \ln x = \infty$, $\lim_{x \rightarrow \infty} x^\pi = \infty$ $\xRightarrow{\frac{\infty}{\infty}}$ L'Hospital rule holds.

$$\therefore \text{LHS} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\pi x^{\pi-1}} = \lim_{x \rightarrow \infty} \frac{1}{\pi x^\pi} = 0 \quad \square$$

• Other indeterminate forms can be transformed into ones that involve

$\frac{0}{0}$ or $\frac{\infty}{\infty}$.

$$(f) \lim_{x \rightarrow \infty} x^2 e^{-\sqrt{x}} = 0. \quad (" \infty \cdot 0 ")$$

$$(g) \lim_{x \rightarrow 0} (1 - 2x)^{3/x} = e^{-6}. \quad ("1^\infty")$$

$$(h) \lim_{x \rightarrow 0^+} \frac{1}{x} - \frac{1}{\sin x} = 0. \quad (" \infty - \infty ")$$

- Apply the L'Hôpital's rule *Carefully* !

Make sure to stop at the right step, or a mistake could arise. For example, the following is wrong:

$$\lim_{x \rightarrow 0} \frac{x^2}{x^2 + \sin x} = \lim_{x \rightarrow 0} \frac{2x}{2x + \cos x} \neq \lim_{x \rightarrow 0} \frac{2}{2 - \sin x} = \frac{2}{2} = 1.$$

$\uparrow$ "0/0" $\parallel$

(The second equality fails because the second limit does not satisfy the assumption of L'Hôpital's rule.)

$$\frac{\lim_{x \rightarrow 0} 2x}{\lim_{x \rightarrow 0} 2x + \cos x} = \frac{0}{1} = 0.$$

- *Limitation* : The rule is not useful for either.

L'Hôpital's rule has its limitation. For example, try to compute the following limits.

$$\textcircled{P} \quad \lim_{x \rightarrow \infty} \frac{x^2 + \sin x}{x^2} \stackrel{\text{"0/0"}}{=} \lim_{x \rightarrow \infty} \frac{2x + \cos x}{2x} \stackrel{L'}{=} \lim_{x \rightarrow \infty} \frac{2 - \sin x}{2}$$

► $\lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$\textcircled{1} \quad LHS = \lim_{x \rightarrow \infty} \left(1 + \frac{\sin x}{x^2} \right) = 1.$$

$\textcircled{2}$

Limits of products and quotients

Suppose that $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L \in \mathbb{R}$. Then

$$\begin{aligned} \lim_{x \rightarrow a} f(x) h(x) &= \lim_{x \rightarrow a} \frac{f(x)}{g(x)} \cdot g(x) \cdot h(x) \stackrel{\text{limit laws}}{=} \lim_{x \rightarrow a} \frac{f(x)}{g(x)} \cdot \lim_{x \rightarrow a} (g(x) h(x)) \\ &= L \cdot \lim_{x \rightarrow a} (g(x) h(x)) \end{aligned}$$

provided the limits on the right-hand side exist.

Rmks : • Replacing $f(x)$ with $L \cdot g(x)$ in a product (or quotient) will keep the limit unchanged, if it exists.

• It does not work with sums and differences in general.

• $\lim_{x \rightarrow a}$ can be replaced with $\lim_{x \rightarrow \infty}$ or $\lim_{x \rightarrow -\infty}$.

• If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1$, we may write $f(x) \sim g(x)$ as $x \rightarrow a$.

e.g. For $f(x) = \sin x$, $\tan x$, $e^x - 1$ and $\ln(1+x)$,

We have $f(x) \sim x$ as $x \rightarrow 0$.

• $\sin x \sim x$ as $x \rightarrow 0$, since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

• $\tan x \sim x$ as $x \rightarrow 0$, since $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{\cos x} \cdot \frac{1}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1 \cdot 1 = 1$.

• $e^x - 1 \sim x$ as $x \rightarrow 0$, since $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \stackrel{L'}{=} \lim_{x \rightarrow 0} \frac{e^x}{1} = e^0 = 1$.

• $\ln(1+x) \sim x$ as $x \rightarrow 0$, since $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \stackrel{L'}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1} = 1$.

e.g. $1 - \cos x \sim \frac{1}{2}x^2$ as $x \rightarrow 0$, since $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\frac{1}{2}x^2} = \lim_{x \rightarrow 0} \frac{2\sin^2 \frac{x}{2}}{\frac{1}{2}x^2} \stackrel{y = \frac{x}{2}}{=} \lim_{y \rightarrow 0} \frac{2\sin^2 y}{\frac{1}{2}(2y)^2} = \lim_{y \rightarrow 0} \frac{2\sin^2 y}{2y^2} = \lim_{y \rightarrow 0} \left(\frac{\sin y}{y}\right)^2 = 1$.

e.g. compute $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = ?$.

Sol': LHS = $\lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - \sin x}{x^3}$

$$= \lim_{x \rightarrow 0} \frac{\sin x \left(\frac{1}{\cos x} - 1\right)}{x^3} = \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{x^3 \cos x}$$

$$\stackrel{\star}{=} \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \cdot \frac{1 - \cos x}{\frac{1}{2}x^2} \cdot \frac{\frac{1}{2}x^2 \cdot x}{x^3} \cdot \frac{1}{\cos x} \right]$$

$$\stackrel{\text{limit law}}{=} \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \cdot \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{\frac{1}{2}x^2} \right) \cdot \frac{1}{2} \left(\lim_{x \rightarrow 0} \frac{1}{\cos x} \right)$$

$$= 1 \cdot 1 \cdot \frac{1}{2} \cdot 1$$

$$= \frac{1}{2}$$

Q: If you replace $\tan x$ and $\sin x$ with x above, you will get

???

$$\text{LHS} \neq \lim_{x \rightarrow 0} \frac{x - x}{x^3} = \lim_{x \rightarrow 0} \frac{0}{x^3} = 0$$

What is the logical error?

A: Limits of products & quotients do NOT work with sums and differences in general!

Relative rates of growth

Consider $P(t)$ being the population at time t . Intuitively, we may say that in long run $P(t) = e^t$ grows faster than $P(t) = \underline{t}$, which grows faster than $P(t) = \ln t$. But we may say that $P(t) = \underline{t^2}$ and $P(t) = \underline{kt^2}$ ($k > 0$) belong to the same class (quadratic growth).

Def: Let f & g be functions which are positive for all sufficiently large inputs. We say that as $x \rightarrow \infty$, f grows

(i) faster than g ; (ii) slower than g ;

(iii) at the same rate as g ;

if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$:

(i) $= \infty$;

(ii) $= 0$;

(iii) $= L \in (0, \infty)$.

Rmk: Usually, f and g above are nondecreasing,
but they do not have to be.

e.g. (a) If f is eventually positive and $k \in \mathbb{R}_+$ is fixed,

then

$$\lim_{x \rightarrow \infty} \frac{k f(x)}{f(x)} = k \in (0, \infty),$$

So $k \cdot f$ and f grow at the same rate.

(b) x^x vs b^x : For any fixed $b > 1$,

$$\lim_{x \rightarrow \infty} \frac{x^x}{b^x} = \lim_{x \rightarrow \infty} \frac{e^{x \ln x}}{e^{x \ln b}} = \lim_{x \rightarrow \infty} e^{x(\ln x - \ln b)} = \infty,$$

So x^x grows faster than b^x .

(c) b^x vs a^x : If $b > a > 1$, then

$$\lim_{x \rightarrow \infty} \frac{b^x}{a^x} = \lim_{x \rightarrow \infty} \underbrace{\left(\frac{b}{a}\right)^x}_{> 1} = \infty,$$

So b^x grows faster than a^x if $b > a > 1$.

(d) a^x vs x^n ($a > 1$, $n \in \mathbb{Z}_+ \triangleq \{1, 2, 3, \dots\}$):

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{a^x}{x^n} &\stackrel{\text{L'H rule}}{=} \lim_{x \rightarrow \infty} \frac{a^x \cdot \ln a}{n \cdot x^{n-1}} \stackrel{\text{L'H rule}}{=} \lim_{x \rightarrow \infty} \frac{a^x \cdot \ln a \cdot \ln a}{n(n-1)x^{n-2}} \\ &\stackrel{\text{L'H rule}}{=} \dots = \lim_{x \rightarrow \infty} \frac{a^x \cdot (\ln a)^{n-1}}{n \cdot (n-1) \dots 2 x^1} \\ &\stackrel{\text{L'H rule}}{=} \lim_{x \rightarrow \infty} \frac{a^x (\ln a)^n}{n!} = \infty, \end{aligned}$$

so a^x grows faster than x^n for any fixed $a \in (1, \infty)$

and $n \in \mathbb{Z}_+$.

Similarly, one can show that a^x grows faster than $\underline{x^r}$ for

any fixed $r > 0$. The proof is left as an exercise.

(e) x^n vs $\ln x$ ($n \in \mathbb{Z}_+$):

$$\lim_{x \rightarrow \infty} \frac{x^n}{\ln x} \stackrel{\text{L'rule}}{=} \lim_{x \rightarrow \infty} \frac{n x^{n-1}}{\frac{1}{x}} = \lim_{x \rightarrow \infty} n \cdot x^n = \infty,$$

so x^n grows faster than $\ln x$, for any fixed $n \in \mathbb{Z}_+$.

The statement is still true if n is replaced by any $r > 0$.

(f) $\log_a x$ vs $\log_b x$ ($a > 1$ and $b > 1$)

$$\lim_{x \rightarrow \infty} \frac{\log_a x}{\log_b x} = \lim_{x \rightarrow \infty} \frac{\frac{\ln x}{\ln a}}{\frac{\ln x}{\ln b}} = \lim_{x \rightarrow \infty} \frac{\ln b}{\ln a} = \frac{\ln b}{\ln a} \in (0, \infty),$$

so log functions with base > 1 all grow at the same rate.

(g) $\log_a x$ vs constant function $g(x) = k$ ($a > 1, k > 0$):

$$\lim_{x \rightarrow \infty} \frac{\log_a x}{k} = \lim_{x \rightarrow \infty} \frac{\frac{\ln x}{\ln a}}{k} = \infty,$$

so any log function with base > 1 grows faster than a const function.

Hence, as $x \rightarrow \infty$, the following list orders the functions from fast to slow growth rates:

① x^x ;

② a^x , $a > 1$ (exponential) ;

③ x^r , $r > 0$ (power) ; $x^{0.01}$, $x^{0.001}$

④ $\log_a x$, $a > 1$ (log) ; $\approx$ x^0 $\neq 1$

⑤ k , $k > 0$ (constant) . $= x^0 = 1$

• Growing at the same rate is a transitive relation !

If f and g grow at the same rate, and g and h also grow at the same rate, then so do f and h , since

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L_1 > 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{g(x)}{h(x)} = L_2 > 0$$

(where L_1, L_2 are finite)

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{f(x)}{h(x)} = \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} \cdot \frac{g(x)}{h(x)} \stackrel{\text{limit laws}}{=} \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} \lim_{x \rightarrow \infty} \frac{g(x)}{h(x)} = L_1 L_2 > 0$$

$\Rightarrow f$ and h grow at the same rate !

finite $\boxed{9}$

e.g. $\sqrt{x^2+2023} \approx x$ and $(98\sqrt{x}-1)^2 \approx x$ grow at the same rate.

Since they both grow at the same rate as x :

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+2023}}{x} = \lim_{x \rightarrow \infty} \sqrt{\frac{x^2+2023}{x^2}} = \lim_{x \rightarrow \infty} \sqrt{1+\frac{2023}{x^2}} = 1$$

$$\lim_{x \rightarrow \infty} \frac{(98\sqrt{x}-1)^2}{x} = \lim_{x \rightarrow \infty} \frac{98^2 x + 1 - 2 \cdot 98\sqrt{x}}{x} = 98^2$$

Similar ideas can be used to talk about the rates at which the functions approach 0:

e.g. • $\lim_{x \rightarrow 0} \sin x = \lim_{x \rightarrow 0} x = 0$, $\lim_{x \rightarrow 0} \frac{\overset{f(x)}{\sin x}}{\underset{g(x)}{x}} = 1$

$\Rightarrow f(x) = \sin x$ and $g(x) = x$ approaches 0 at the same rate as $\boxed{x \rightarrow 0}$.

• $\lim_{x \rightarrow 2} \frac{\overset{f(x)}{(x-2)^2}}{\underset{g(x)}{x-2}} = \lim_{x \rightarrow 2} (x-2) = 0$

$\Rightarrow f(x) = (x-2)^2$ approaches 0 faster than $g(x) = x-2$, as $\boxed{x \rightarrow 2}$.

Big-Oh and Little-Oh

Def : Let f and g be functions, both positive for sufficiently large x . Then we write :

$$f(x) = o(g(x)) \text{ as } x \rightarrow \infty \text{ if } \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0. \quad (1)$$

$$f(x) = O(g(x)) \text{ as } x \rightarrow \infty \text{ if } \exists N, M \in \mathbb{R}, \text{ such that}$$

$$\frac{f(x)}{g(x)} \leq M \quad \forall x \in [N, \infty). \quad (2)$$

$$\Leftrightarrow f(x) \leq M \cdot g(x) \quad \forall x \in [N, \infty).$$

Rmks

• These are called little-Oh and big-Oh notation, respectively.

• (1) states "growth rate of f $<$ growth rate of g "
" f grows slower than g as $x \rightarrow \infty$ ", while

(2) states " f grows at most as fast as g , as $x \rightarrow \infty$ ".

"growth rate of f $\leq$ growth rate of g "

• If $f(x) = o(g(x))$ as $x \rightarrow \infty$, then $f(x) = O(g(x))$ as $x \rightarrow \infty$.

(Why?)

• If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = k \in \mathbb{R}_{>0}$, then $f(x) = O(g(x))$ as $x \rightarrow \infty$.

(Why?) "growth rate of f $\approx$ growth rate of g "

e.g. (a) $k = \underline{O(1)}$. ← constant function

(b) More generally,

$k f(x) = O(f(x))$, if f is eventually positive.

(c) $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = O(x^n)$ if $a_n > 0$, since

$$\lim_{x \rightarrow \infty} \frac{\sum_{i=0}^n a_i x^i}{x^n} = \lim_{x \rightarrow \infty} \sum_{i=0}^n a_i \underbrace{x^{i-n}}_{\rightarrow 0 \text{ unless } i=n} = a_n + a_{n-1} \frac{1}{x} + a_{n-2} \frac{1}{x^2} + \dots + a_0 \frac{1}{x^n}$$

showing that $\sum_{i=0}^n a_i x^i$ and x^n have the same growth rate.

(d) $x + \sin x = O(x)$, since

$$\lim_{x \rightarrow \infty} \frac{x + \sin x}{x} = \lim_{x \rightarrow \infty} \left(1 + \frac{\sin x}{x}\right) = 1,$$

showing that $x + \sin x$ and x have the same growth rate.

(e) $\ln(\ln x) = o(\ln x)$, since

$$\lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{\ln x} \stackrel{y = \ln x}{=} \lim_{y \rightarrow \infty} \frac{\ln y}{y} \stackrel{\text{L'rule}}{=} \lim_{y \rightarrow \infty} \frac{\frac{1}{y}}{1} = 0,$$

showing that $\ln(\ln x)$ grows slower than $\ln x$ as $x \rightarrow \infty$.

Theorem (L'Hôpital's Rule, or L'Hospital's Rule)

Let $c \in \mathbb{R}$. Suppose that f and g are differentiable on $D := (c - a, c + a) \setminus \{c\}$ for some $a > 0$, and that $g'(x) \neq 0$ for all $x \in D$. Suppose that one of the following two conditions holds:

- ▶ $\lim_{x \rightarrow c} f(x) = 0 = \lim_{x \rightarrow c} g(x)$; ③ $\frac{0}{0}$
- ▶ $\lim_{x \rightarrow c} f(x) \in \{-\infty, \infty\}$ and $\lim_{x \rightarrow c} g(x) \in \{-\infty, \infty\}$. $\frac{\pm \infty}{\pm \infty}$ ☆

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)},$$

provided that ④ the limit on the right exists (or is ∞ or $-\infty$). ✓

Proof of the L'Hospital's Rule (Version in the Book) (Optional)

Here we aim to prove one case of the L'Hospital rule: assume in the statement of the L'Hospital's rule that:

- f, g differentiable on $(c-a, c+a)$, $g' \neq 0$ on $(c-a, c+a) \setminus \{c\}$;
- $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$; $\left(\frac{0}{0}\right)$ " $\frac{0}{0}$ "
- $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$. ($L \neq \pm\infty$)

For other cases, proofs are omitted. We need to use and first prove Cauchy's Mean Value Theorem.

Thm (Cauchy). Suppose that f and g are continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

provided that $g(a) \neq g(b)$ and g' is never zero.

- Recall Lagrange's Mean Value theorem: g :
 f : cont' on $[a, b]$, diff' on (a, b)

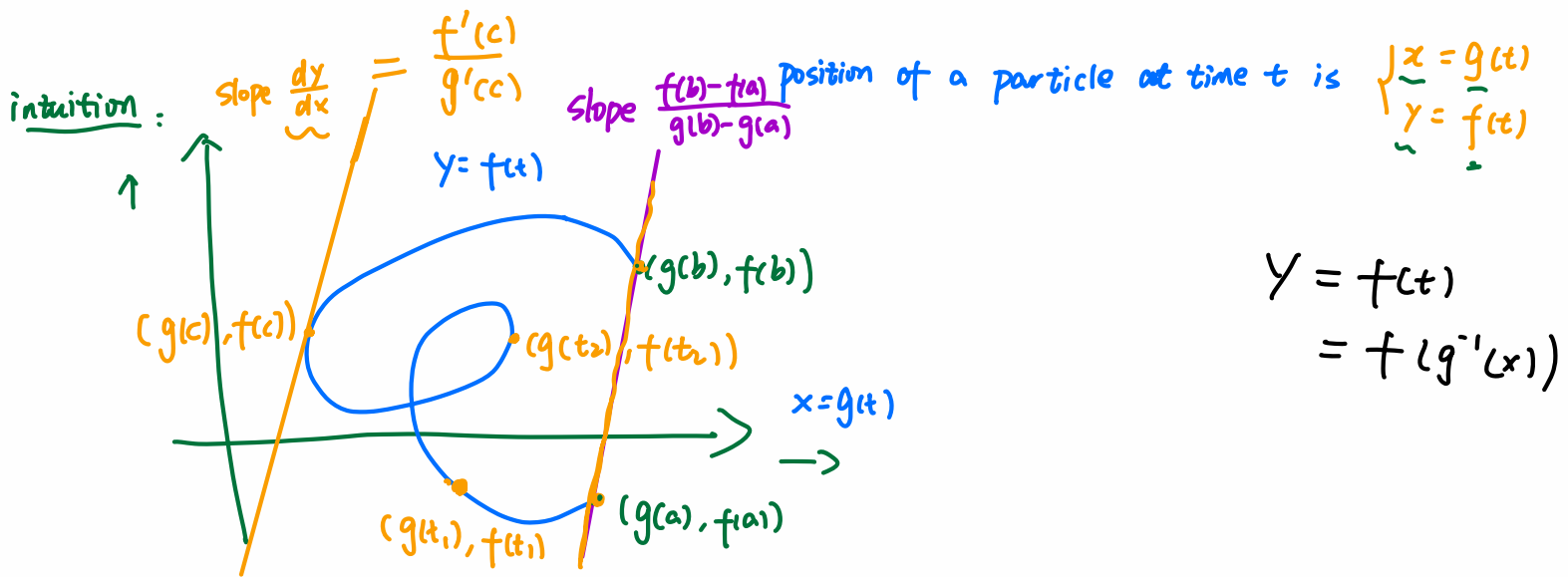
$$(1) \quad f'(c_1) = \frac{f(b) - f(a)}{b - a}, \quad \exists c_1 \in (a, b);$$

$$(2) \quad g'(c_2) = \frac{g(b) - g(a)}{b - a}, \quad \exists c_2 \in (a, b).$$

$$\frac{f'(c_1)}{g'(c_2)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$c_1, c_2 \in (a, b)$

- It is a stronger version of Lagrange's MVT (Let $g(x) = x$).



$$\textcircled{1} \quad \left. \frac{dy}{dx} \right|_{t=c} = \left. \frac{dy/dt}{dx/dt} \right|_{t=c} = \left. \frac{df(t)/dt}{dg(t)/dt} \right|_{t=c} = \frac{f'(c)}{g'(c)}$$

$$\textcircled{2} \quad \left. \frac{dy}{dx} \right|_{t=c} \xrightarrow[\text{exists}]{\text{assume } g^{-1}} \left. \frac{d f(g^{-1}(x))}{dx} \right|_{t=c} = f'(g^{-1}(x)) \left. \frac{1}{g'(g^{-1}(x))} \right|_{t=c} (g^{-1}(x)=t) \\ = f'(t) \cdot \frac{1}{g'(t)}$$

Pf: Suppose that $g(a) \neq g(b)$. Define h on $[a, b]$ by

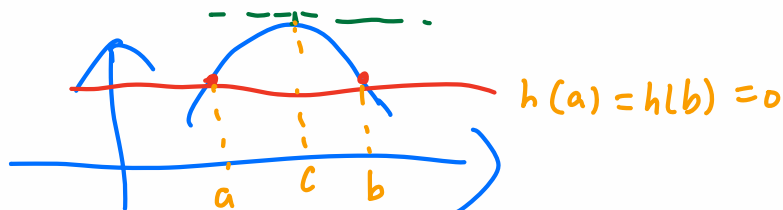
$$h(t) \triangleq f(t) - \left[f(a) + \frac{f(b)-f(a)}{g(b)-g(a)} (g(t)-g(a)) \right]$$

• Then h is cont on $[a, b]$ and diff on (a, b) .

• Note that $h(a) = f(a) - \left[f(a) + \frac{f(b)-f(a)}{g(b)-g(a)} (g(a)-g(a)) \right] = 0$

and $h(b) = f(b) - \left[f(a) + \frac{f(b)-f(a)}{g(b)-g(a)} (g(b)-g(a)) \right] = 0$.

• By Rolle's Thm, $\exists c \in (a, b), h'(c) = 0$.



• Since $h'(t) = f'(t) - \frac{f(b)-f(a)}{g(b)-g(a)} g'(t)$, $h'(c) = 0$ and $g'(c) \neq 0$,

we have $h'(c) = 0 \Rightarrow f'(c) - \frac{f(b)-f(a)}{g(b)-g(a)} g'(c) = 0$

$$\Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)} .$$

[4]

Proof of L'Hospital's rule ($\frac{0}{0}$) (Optional)

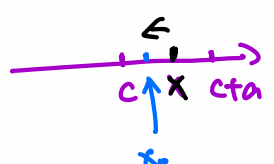
Step 1: right limit!

• First show that it holds for $\lim_{x \rightarrow c+}$.

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

Step 2: Cauchy's MVT!

• Pick any $x \in (c, c+a)$. By Cauchy's MVT, $\exists x_0 \in (c, x)$, s.t. $\frac{f(x)-f(c)}{g(x)-g(c)} = \frac{f'(x_0)}{g'(x_0)}$

$$\frac{f'(x_0)}{g'(x_0)} = \frac{f(x)-f(c)}{g(x)-g(c)} \quad \frac{0}{0} \quad (*)$$


[For Cauchy's MVT to be valid, one needs to check:

$g(x) \neq g(c)$ and $g'(x)$ is never zero.

Indeed, in the first assumption of L'Hospital's rule, it says that

• f, g differentiable on $(c-a, c+a)$, $g' \neq 0$ on $(c-a, c+a) \setminus \{c\}$;

Hence, g' is never zero on $(c, c+a)$ (hence on (c, x));

and $g(x) \neq g(c)$ (otherwise ^{there} exists some $c_0 \in (c, x)$ s.t. $g'(c_0) = 0$ which is a contradiction).

]

Step 3 : compute !

Since f and g are both differentiable at c , they are continuous at c , so

$$\lim_{x \rightarrow c} f(x) \stackrel{\text{cont}}{=} f(c) \stackrel{\frac{0}{0}}{=} 0,$$

[due to $\frac{0}{0}$]

$$\lim_{x \rightarrow c} g(x) \stackrel{\text{cont}}{=} g(c) \stackrel{\frac{0}{0}}{=} 0.$$

From (*), we have

$$\underbrace{\frac{f(x)}{g(x)}}_{x \rightarrow c+} = \frac{\underbrace{f(x) - f(c)}_{x \rightarrow c+} \underbrace{f'(x_0)}_{x \rightarrow c+}}{\underbrace{g(x) - g(c)}_{x \rightarrow c+} \underbrace{g'(x_0)}_{x \rightarrow c+}}$$

$$\begin{array}{ccc} c < x_0 < x \\ \downarrow & & \downarrow \\ c- & & c+ \end{array}$$

As $x \rightarrow c+$, we have $x_0 \rightarrow c+$ and

$$\lim_{x \rightarrow c+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c+} \frac{f'(x_0)}{g'(x_0)} = \lim_{x_0 \rightarrow c+} \frac{f'(x_0)}{g'(x_0)}.$$

Hence,

$$\lim_{x \rightarrow c+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c+} \frac{f'(x)}{g'(x)}.$$

Step 4. left limit !

The case for $\lim_{x \rightarrow c-}$ is symmetric.

□